Keep the Clock, Contrast the Prior: Training-Free Temporal Grounding for Video LLMs under a Visual-Token Budget

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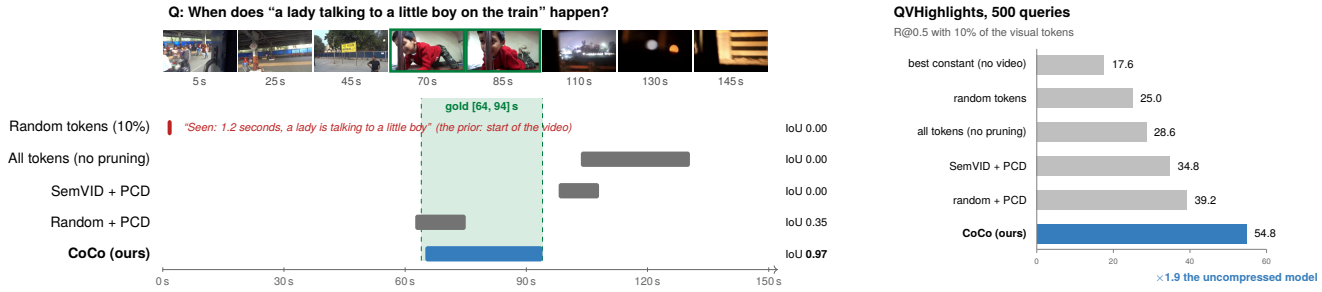


Figure 1. **Temporal grounding from a pruned video memory.** Left: a QVHighlights query, with the intervals predicted by each method at 10% of the visual tokens (real outputs). With random pruning the model falls back on a video-free prior and answers "1.2 seconds"; QA-oriented pruning and the uncompressed model pick a later, wrong scene. CoCo keeps every frame anchored in time and contrasts the answer against the model's no-video prediction, and recovers the interval (IoU 0.97). Right: over 500 queries, CoCo nearly doubles the R@0.5 of the uncompressed model and is far above the best video-free constant.

Abstract

Video LLMs are increasingly asked not only what *happened but* when*: a grounded question such as "how long did I use the paper cutter?" needs an answer and the time interval of its evidence. To keep long videos tractable, the visual memory of these models is pruned to a small fraction of its tokens, and published pruners lose little multiple-choice accuracy at 10% of the tokens. We show that temporal grounding behaves differently. Four recent training-free pruners fall* below random token selection *on grounding, because a model reads time from the timestamps that survive in memory, and selection by saliency, uniqueness or text similarity removes whole stretches of the timeline. Where the memory is silent, the decoder fills the gap with a prior that does not depend on the video: asked with no video at all, the model still answers, starting intervals early and ending them too soon. We propose CoCo, a training-free method with two parts.* Coverage *gives every frame an equal token quota and fills it with the tokens the actual question attends to, so that every time anchor survives pruning.* Prior-contrastive decoding *(PCD) writes each time value against the model's own no-video prediction, with a strength that fades as the model grows confident, and reserves room for the final interval. Because grounding benchmarks carry strong positional biases, we evaluate every method against video-free floors. With 10% of the visual tokens, CoCo raises R@0.5 on QVHighlights from 25.0 (random tokens) and 28.6 (all tokens) to* ***54.8****, where the best video-free constant reaches 17.6, and from 24.6 to* ***40.8*** *on S-EMBER grounded QA. We also report where it does not help: on Charades-STA and ActivityNet Captions the uncompressed model itself stays below the video-free floor, and no pruning method, ours included, rises above it.*

1. Introduction

Ask a video assistant "how long did I use the paper cutter?" and a useful answer has two parts: a duration and the interval it comes from. Such *grounded* questions are central to egocentric assistants, video search and step-by-step guidance. They are hard for the same reason they are useful: the model must know *when* things happened across the whole window, including the moments just before and after the event.

At the same time, video LLMs pay for every frame. Qwen3-VL [32] spends about 330 visual tokens per pair

of frames, so 64 frames already cost about 21,000 tokens, and long or streaming video multiplies this. Pruning the visual tokens to a small budget is therefore the default, and a fast-growing literature reports that 10–25% of the tokens preserve multiple-choice accuracy [15, 18, 23, 30, 33, 35]. This paper asks what pruning does to *temporal grounding*, and finds that the answer differs from QA in three ways.

Observation 1: QA pruners lose the timeline. Video LLMs such as Qwen3-VL report time by reading timestamp tokens interleaved with the visual tokens. A frame whose visual tokens are all removed loses its timestamp, or keeps a timestamp with nothing to show. Pruners designed for QA remove frames that look redundant, unremarkable or unrelated to the words of the question, and those frames are exactly what makes a timeline usable for grounding. On S-EMBER grounded QA, all four recent pruners we evaluate fall below *random* token selection at the same budget, and the same holds on QVHighlights and Charades-STA (Sec. 6.2). Random selection is a strong baseline for grounding because it keeps a few tokens on every frame.

Observation 2: where the memory is silent, a prior answers. Asked with no video at all, the model still produces confident intervals, and 90% of them start in the first 5% of the window. A pruned memory shows the same habit: 30–56% of predicted intervals start in the first 5% of the window, against 16% of the gold intervals, and removing the frames that contain the evidence raises this share from 37% to 51%. The end of the interval suffers most: the median end error is two to three times the start error, and without video 87% of predicted ends fall more than half an interval too early (Sec. 3.3).

Observation 3: benchmark floors are high. Grounding benchmarks have strong positional regularities [31]. On S-EMBER, where questions are asked mid-stream and two thirds of the gold intervals end within the last fifth of the window, the best *constant* interval $[0.15q, q]$ reaches 36.9 mIoU, more than the uncompressed model (28.0). On ActivityNet Captions and Charades-STA the best constant also beats the uncompressed model. A claimed gain in grounding under compression must clear these floors, and we report them for every method (Sec. 3.4).

These observations suggest that a compressed memory for grounding needs two things that QA-oriented pruning does not provide: **coverage**, so that every part of the timeline keeps an anchor in memory, and a way to stop the decoder's **prior** from writing the interval where the evidence is thin. We propose CoCo (Coverage + PCD), a training-free method with one component for each (Fig. 3):

- **Coverage** (Sec. 4.1). After the frames are encoded, the KV cache is pruned layer by layer. Every frame keeps an equal share of the layer's budget, and within each frame the tokens that the *actual question* attends to most are kept. Because a timestamp survives exactly when one of its frame's visual tokens survives, the equal quota keeps every time anchor, and the question decides what each anchor shows.
- **Prior-contrastive decoding (PCD)** (Sec. 4.2). While the answer is written, each time value is chosen by contrasting the model's prediction with its prediction from a video-free memory. PCD acts only on time values, and its strength fades where the model is already confident. An *answer reserve* (Sec. 4.2), part of PCD, guarantees that the final interval is always written.

Neither operator is new in isolation: per-frame budgets appear in VidCom^2, KTV and AdaTP [2, 13, 30], question-attention KV eviction in SnapKV and PruneVid [23, 27], and contrast against a degraded input in VCD and M3ID [19, 26]. Our contributions are to show *why* temporal grounding needs this particular combination, and to make it work:

1. **A diagnosis of grounding under compression** (Sec. 3). QA-oriented pruners lose temporal coverage and fall below random selection; the decoder's video-free prior fills the gaps, mostly at the interval end; and benchmark floors are high enough to hide both effects.
2. **CoCo** (Sec. 4), which couples time-anchor coverage with prior-contrastive decoding of time values. It needs no training, no labelled calibration and no change to the model.
3. **A floor-adjusted evaluation** (Sec. 5) that reports, for every method, the best video-free constant interval, a length-rescaling baseline, and closed versus open intervals, with paired confidence intervals. All design choices were made on S-EMBER; the other three datasets were run once with the frozen configuration.
4. **Results on four datasets** (Sec. 6). With 10% of the tokens, CoCo exceeds the uncompressed model and every pruner on S-EMBER and QVHighlights, and clears the video-free floors by $+7.6$ mIoU on S-EMBER closed intervals (appendix) and $+22.1$ on QVHighlights, both significant. Question-driven selection improves as the budget shrinks, as long as each frame keeps enough tokens; at very small budgets, fewer frames with more tokens each recover what 64 frames lose. We also report the two datasets where the base model itself stays below the floor, and a broad set of negative results.

2. Related Work

Frame selection. BOLT [29] scores frames against the query with a contrastive image–text model and selects them by inverse transform sampling; AKS [34], MDP3 [6], FOCUS [3] and Q-Frame [7] optimise relevance together with coverage or diversity. All are evaluated on multiple-choice QA. Our optional retrieval front end (Sec. 4.3) is BOLT adapted to grounding, and we find that it contributes little once tokens are chosen by the question.

Visual token pruning for video LLMs. Query-agnostic methods compress by redundancy or saliency: FastVID [33] segments the video and keeps density peaks, VidCom^2 [30] gives each frame a budget by its uniqueness, FlashVID [18] selects by [CLS] attention and merges tokens across frames with trees, and DyCoke [35] and FrameFusion [21] merge similar tokens. Query-aware methods use the text: SemVID [15] scores tokens by word-embedding relevance with MMR diversity, AdaTP [2] boosts segments similar to the text, LGTTP [5] reads temporal words in the question to reallocate tokens, and PruneVid [23] keeps the top 40% of tokens by question attention at one LLM layer. KTV [13] is closest in structure (keyframes, then tokens with a relevance-weighted per-frame budget), and LGTTP is the only method we know that evaluates grounding, at a milder budget and offline. CoCo differs in three ways: the per-frame quota is equal by design, the score is the actual question's attention propagated exactly through all layers, and the quota exists to keep every timestamp, which we show matters for grounding.

KV-cache compression and video memory. SnapKV [27], H_2O [39], PyramidKV [16] and AdaKV [20] evict KV entries by attention in long-context LLMs; our token stage is SnapKV-style eviction applied to video with a per-frame constraint. For streaming video, HERMES [11] compresses the cache with generic pseudo-queries, ReKV [17] retrieves KV blocks at question time, and StreamMem [9] and InfiniPot-V [4] maintain bounded memories. These works report QA only. Our memory is built per question from the frames up to the question time: it is causal, but it is not a persistent streaming memory, and we make no claim about the cost of building it (Sec. 7).

Contrastive decoding. VCD [26] contrasts the output distribution against a noised image, ICD [37] against a disturbed instruction, M3ID [19] against the absence of the image, and PAI [28] amplifies image attention while subtracting text-only logits; SEASON [8] extends the idea to video with temporally corrupted negatives. These methods target object hallucination or QA. We apply the no-input contrast to *time values only*, with a confidence-adaptive strength, and test whether its benefit for grounding is more than a uniform lengthening of the interval (Sec. 6.4).

Training-free temporal grounding. NumPro [38] draws frame numbers on the frames, FV-Action [10] scans the video with yes/no questions, REZE [14] recasts grounding as recognition, and [Authors to verify] [12] read intervals from calibrated attention heads, debiased against a blank-video run. All assume the full set of video tokens. CoCo addresses the complementary problem of grounding from a *compressed* memory; these readouts could run on top of it **[TODO: run FV-Action and the attention readout on the same 10% memory]**.

Benchmark biases. Otani et al. [31] showed that blind baselines are competitive on Charades-STA and ActivityNet Captions because of positional priors in the annotations. We find the same on the datasets we use, and make video-free floors part of every comparison.

3. Grounding under a Token Budget: a Diagnosis

This section establishes the three observations of the introduction on S-EMBER, before any method is introduced. Unless stated otherwise, numbers in this section use random 10% pruning as the reference memory.

3.1. Setting

A question x is asked at time q about a video $v_{0:q}$. The model sees F=64 frames from $[0, q]$, encoded into a KV cache $\mathcal{M}$ of N visual tokens interleaved with timestamp tokens (one "`<t seconds>`" per pair of frames in Qwen3-VL). A budget ρ (default ρ=0.1) limits the number of tokens kept in each layer to ρN, with kept timestamps counted inside the budget. The model answers with a short *timeline*:

```
Seen: t_1 seconds, …
Seen: t_k seconds, …
Answer: …
Time: [s, e]
```

from which the interval $[s, e]$ is parsed. Grounding is scored by its IoU with the gold interval (mIoU and R@$\{0.3, 0.5, 0.7\}$); grounded QA additionally requires the answer to be judged correct (GQ@0.5: correct answer and IoU ≥ 0.5). We call an answer an *early start* when its interval starts in the first 5% of the window, and a *leak* when it starts early although the gold evidence starts later.

3.2. QA pruners lose temporal coverage

At 10% of the tokens, FastVID, VidCom^2, FlashVID and SemVID all score below random token selection (Sec. 6.2). The cause is visible in where the kept tokens lie (Tab. 1). HERMES's attention-plus-recency score keeps 27.6% of the tokens in the last quarter of the window and only 2.5–4.7% elsewhere; FastVID keeps a token from only 25% of the sampled frames in a median question, because it retains one representative per run of similar frames, and runs of similar frames are exactly where a continuing event lives. Random selection keeps about 9% everywhere and never empties a frame. A timestamp is useful only when visual content of its frame survives with it, so a selector that empties frames leaves holes in the timeline. Holes hurt even when the score itself is reasonable: ranking HERMES's scores within each frame (an equal per-frame share, "stratified") instead of globally gains 5.3* mIoU.

Table 1. **Where pruning keeps tokens.** Share of the visual tokens kept, by position in the window (S-EMBER, 10% budget), and the resulting grounding (mIoU) and leak. FastVID (mIoU 17.9, leak 38%) keeps 6.0 / 8.5 / 10.5 / 10.6 / 12.9% over the five fifths of the window, and tokens from only 25% of the frames.

Selector	0–10%	10–25%	25–50%	50–75%	75–100%	mIoU	Leak
HERMES	4.7	3.9	2.6	2.5	27.6	18.3	49%
Stratified	8.7	8.7	8.6	8.6	8.7	23.7	**29%**
Random	8.9	8.8	8.6	8.7	8.6	**25.8**	30%

3.3. The prior fills the gaps

The model answers without video. Asked every question with an empty visual memory, the model still writes a timeline: 89% of its first `Seen:` moments are at 0 s and 90% of its intervals start in the first 5% of the window (88% leak). Its intervals are 0.45× the gold length (median) and 87% of its ends are more than half an interval too early. We call this the model's *temporal prior*: start early, stop soon.

The prior leaks into answers with video. The same signature appears in every memory, weaker (Fig. 2). About 30% of intervals start early at every random budget and with no pruning at all, against 16% of gold intervals, and the share rises when coverage is lost (HERMES 49%, HERMES streaming 37–39%). The leak is therefore not a matter of the number of tokens, but of where the memory is silent.

Removing the evidence causes the leak. To test causality we drop the uniform frames that fall inside the gold interval ("evidence") or the same number of frames elsewhere ("control"), keeping every other frame with its true timestamp (Tab. 2). With the evidence removed, the answers do not spread out: they move to the start of the video, and the leak rises by 14 points for both the unpruned and the random 10% memory. Within the HERMES 10% memory, questions whose evidence keeps under 5% of its tokens leak 56% of the time, against 33% when at least 15% survives.

An instruction does not remove it. Adding "The relevant moments may occur at any point in the video; do not assume they are near the start" to the prompt leaves the leak almost unchanged (29.5% → 28.0%) and grounding unchanged (−0.1 mIoU). The prior acts in the output distribution and has to be removed there.

The end is the weak boundary. Across memories, the median end error is 0.40–0.69 of the gold length without contrastive decoding, against 0.14–0.30 for the start, and the most frequent failure is an interval entirely *before* the gold one (33% of all answers, Tab. 20).

3.4. Benchmark floors

Positional regularities make some intervals predictable without the video (Tab. 3). On S-EMBER, questions are asked while the stream is running and 67% of gold intervals end

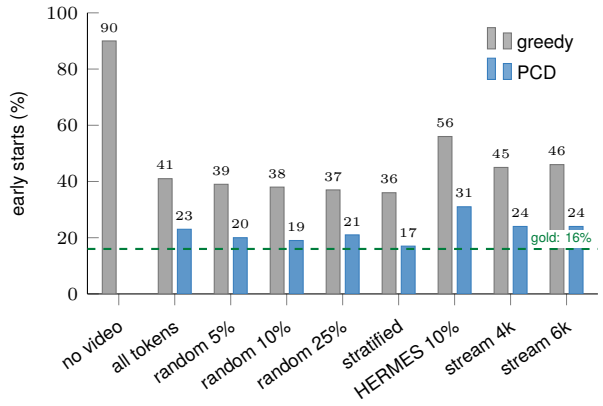


Figure 2. **The temporal prior leaks into every memory, and PCD removes most of it.** Share of predicted intervals that start in the first 5% of the window (S-EMBER). Without video the model starts early 90% of the time; every memory inherits part of this habit, most where pruning loses coverage (HERMES). PCD roughly halves it everywhere.

Table 2. **Causal test.** Frames inside the gold evidence ("evidence") or as many frames elsewhere ("control") are removed. Acc. / mIoU / GQ@0.5 and leak; leak measured on the 397 questions whose evidence starts after the first 5% of the window. PCD: Sec. 4.2; earlier runs: PCD without its answer reserve.

		Greedy		PCD	
Memory	Removed	Acc./mIoU/GQ	Leak	Acc./mIoU/GQ	Leak
Random 10%	control	16.6/28.8/8.8	**31%**	15.4/34.6/10.5	**11%**
	evidence	7.8/ 8.6/1.7	46%	9.3/10.3/1.1	23%
All tokens	control	**19.6/29.4/10.1**	37%	**18.1/35.9/12.6**	15%
	evidence	9.1/ 9.4/1.9	51%	9.1/10.5/1.5	22%

within the last 20% of the window, so the constant $[0.15q, q]$ scores 36.9 mIoU. On ActivityNet Captions, whose moments cover 38% of the video on average, $[0, 0.95q]$ scores 37.9; on Charades-STA, $[0, 0.45q]$ scores 29.7. QVHighlights is the least biased (26.8 mIoU, 17.6 R@0.5). These constants are tuned on the test questions, so they upper-bound any video-free rule of this form. On three of the four datasets the best constant beats the uncompressed model, which makes the floors a necessary reference for every comparison that follows.

4. Method: CoCo

CoCo acts at the two places where Sec. 3 locates the failures: the memory (Coverage) and the decoder (PCD). Neither needs training. Figure 3 gives an overview.

4.1. Coverage: equal quotas filled by the question

After the F frames are encoded, the memory is pruned once, independently in each of the L layers. Let $\mathcal{T}_f$ be the visual

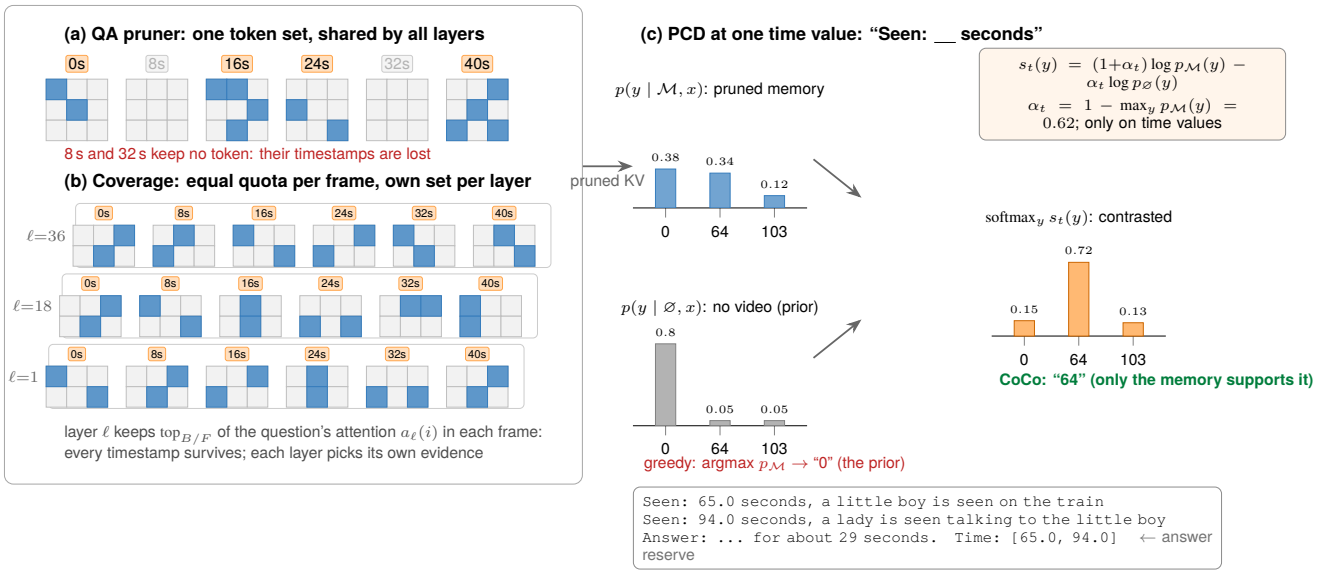


Figure 3. **Overview of CoCo. (a)** QA-oriented pruners keep one set of salient or unique tokens in every layer and can empty whole frames, removing their timestamps. **(b)** Coverage prunes each layer separately: every frame keeps the same share of the layer budget, filled with the tokens that the question attends to most *in that layer*. Every timestamp survives in every layer, and different layers keep different evidence (sharing one set across layers costs 1.5 mIoU, Tab. 10). **(c)** PCD at one time value of the teaser example (probabilities illustrative). Greedy decoding picks "0", which the memory supports only as much as the video-free prior does; the contrasted score discounts it and promotes "64", which only the memory supports. The answer reserve guarantees that the final interval is written (output shown is the real CoCo answer).

Table 3. **Video-free floors.** Best constant interval $[aq, bq]$ on a 20×20 grid, tuned on the test questions (each metric uses its own best constant; $[a, b]$ is the mIoU optimum), against the uncompressed model. Closed: share of questions with gold end $< 0.8q$.

	Best constant					All tokens	
Dataset	$[a, b]$	mIoU	R@0.5	R@0.7	Closed	mIoU	R@0.5
S-EMBER	$[.15, 1]$	36.9	34.7	24.2	33%	28.0	26.7
QVHighlights	$[0, 1]$	26.8	17.6	8.8	62%	29.1	28.6
ActivityNet	$[0, .95]$	37.9	35.8	20.8	56%	22.9	18.2
Charades-STA	$[0, .45]$	29.7	29.6	15.4	70%	16.6	14.2

tokens of frame pair $f \in \{1, \dots, F'\}$ and B the per-layer budget for visual tokens.

Question relevance. The question x is appended to the full, uncompressed memory and propagated exactly through all layers, so that the queries at layer ℓ come from the real hidden states. The relevance of visual token i at layer ℓ is the attention it receives from the question, averaged over heads h and question positions j:

$$a_\ell(i) \;=\; \frac{1}{H\,|x|}\sum_{h=1}^{H}\sum_{j\in x}\mathrm{softmax}_i\Big(\tfrac{1}{\sqrt{d}}\,Q_j^{(\ell,h)}\,K_i^{(\ell,h)^\top}\Big). \tag{1}$$

Equal quota. Every frame pair receives the same share of the budget, $B_f = \lfloor B/F' \rfloor$ (remainders assigned in order), and layer ℓ keeps

$$\mathcal{K}_\ell \;=\; \bigcup_{f=1}^{F'} \mathrm{top}_{B_f}\big\{\, a_\ell(i) \;:\; i \in \mathcal{T}_f \,\big\}. \tag{2}$$

The timestamp tokens of f are kept if and only if $\mathcal{K}_\ell \cap \mathcal{T}_f \neq \emptyset$. With an equal quota every frame pair keeps at least one token at any practical budget, so *every time anchor survives*, and the question decides which part of each frame is shown. Kept tokens retain their original rotary positions, so their temporal order is preserved.

Why equal, and why the question. The equal quota is a deliberately simple coverage constraint. It guarantees the timeline, which grounding needs and multiple-choice QA does not, and it stops attention from collapsing onto a few frames: when quotas follow attention mass, attention sinks and recency bias give a single frame up to 227 of about 2,140 kept tokens. Relevance-weighted quotas with a floor, from SigLIP similarity or attention mass, did not beat equal quotas in a pre-registered test (-0.2 mIoU). Using the *actual* question, rather than the generic probe questions of streaming memories such as HERMES [11], is what makes attention informative: generic probes with a global top-k are the weakest variant we measured (18.3 mIoU).

Per-layer selection. Each layer keeps its own set $\mathcal{K}_\ell$. Forcing one shared set (attention averaged over layers) costs

CoCo -1.5^* mIoU, whereas it does not change random selection ($+0.7$, n.s.). The benefit of per-layer selection is therefore specific to question-driven scores, and it does not explain why random selection beats input-level pruners.

Frames and budget. An equal quota needs enough tokens per frame to show anything: with 2–4 tokens per frame pair, the top-attended tokens carry little content and Coverage collapses (Sec. 6.6). When the budget is small, we therefore spend it on fewer frames: at a fixed number of kept tokens, 32 frames with twice the share per frame recover what 64 frames lose. As a rule we use F=64 when each frame pair keeps at least $\sim$8 tokens and halve F otherwise. **[TODO: this rule was derived from the results of Sec. 6.6; validate it on held-out data.]**

Cost. Coverage adds one forward pass of the question over the uncompressed memory per question. **[TODO: measured latency and peak memory; the current implementation runs 3–4 scoring passes, reducible to one.]**

4.2. PCD: prior-contrastive decoding of time values

Let $p_{\mathcal{M}}(y) = p(y \mid \mathcal{M}, x, y_{<t})$ be the next-token distribution with the compressed memory and $p_{\varnothing}(y) = p(y \mid \varnothing, x, y_{<t})$ the distribution with the same prompt and generated prefix but *no visual tokens*. Then

$$\log p_{\mathcal{M}}(y) \;=\; \log p_{\varnothing}(y) \;+\; \underbrace{\log \frac{p_{\mathcal{M}}(y)}{p_{\varnothing}(y)}}_{\text{evidence from } \mathcal{M}}, \qquad (3)$$

so when the memory carries no information about y, the evidence term is flat and the video-free prior decides: this is the failure of Sec. 3.3. Prior-contrastive decoding (PCD) amplifies the evidence term,

$$\begin{aligned} s_t(y) \;&=\; (1+\alpha_t)\log p_{\mathcal{M}}(y) - \alpha_t \log p_{\varnothing}(y) \\ &=\; \log p_{\varnothing}(y) + (1+\alpha_t)\log \frac{p_{\mathcal{M}}(y)}{p_{\varnothing}(y)}, \end{aligned} \qquad (4)$$

and picks $y_t = \arg\max_{y \in \mathcal{V}_\beta} s_t(y)$ over the plausible set $\mathcal{V}_\beta = \{y : p_{\mathcal{M}}(y) \geq \beta \max_{y'} p_{\mathcal{M}}(y')\}$ with β=0.1, as in VCD [26], so that tokens the memory makes implausible are never promoted. Three choices adapt this rule to grounding.

No-video counterfactual. The negative memory is empty rather than noised or shuffled. It isolates exactly the prior measured in Sec. 3.3, and it is cheap: its cache holds only the prompt.

Time scope. PCD is applied only while a time value is written: inside a `Seen:` time before its description and inside `Time: [s, e]`. All other tokens, including the answer text, are decoded greedily from $p_{\mathcal{M}}$, so the answer is unaffected where the memory already speaks.

Confidence-adaptive strength. We set $\alpha_t = \alpha_{\max}\big(1 - \max_y p_{\mathcal{M}}(y)\big)$ with $\alpha_{\max}$=1. Where the memory makes the model confident (*e.g.* evidence that truly starts at 0 s), PCD fades and the early answer is kept; where the model hesitates between times, the evidence term decides.

The timeline prompt matters: PCD acts on the listed `Seen:` moments and then on the interval, so the interval is written from contrasted observations. **[TODO: report PCD with a direct-answer prompt, where it does not help.]**

Answer reserve. The last part of PCD is a guard on the output format. Contrastive decoding lengthens the list of observed moments: under PCD the model sometimes keeps writing `Seen:` lines until the generation limit T, and never writes the interval. We reserve room for the answer: if R tokens remain (R=64) and no `Answer:` line has started, the decoder writes "`\nAnswer:`" and continues normally. Throughout the paper, PCD includes this reserve. This is a format constraint, not a heuristic on the content: the interval is still chosen by PCD from the memory. It removes every unparseable output (from 25–68 of 500 to 0) and improves every configuration on every dataset (Tab. 15).

4.3. Optional: retrieval-guided frames

When the window is much longer than F frames, which frames are encoded can matter as much as which tokens are kept. As an optional front end we sample frames as in BOLT [29]. Each second of video is embedded once with SigLIP 2 [36], which gives a causal 1 fps index. The relevance z_j of second $j \leq q$ is its z-scored cosine similarity with the question, and the F frames are taken at evenly spaced quantiles of

$$p(j) \;=\; \tfrac{1}{2}\cdot\tfrac{1}{q} \;+\; \tfrac{1}{2}\,\mathrm{softmax}(z)_j, \qquad (5)$$

so that half of the frames stay uniform. Retrieval is skipped when uniform frames would be less than 1 s apart. Once tokens are chosen by the question, retrieval adds only $+0.8$ to $+0.9$ mIoU (not significant on S-EMBER, QVHighlights or ActivityNet), so we treat it as optional; the main-table runs include it.

5. Experimental Setup

Model. Qwen3-VL-8B-Instruct [32], 64 frames sampled uniformly from $[0, q]$ (or by Sec. 4.3) at native resolution, greedy decoding or PCD, at most 384 new tokens, batch size 8. All methods share the timeline prompt.

Datasets. **S-EMBER** grounded QA [1]: 475 questions (191 duration, 184 counting, 100 location) on the first 300 videos, asked mid-stream; the memory contains only the frames before the question. **QVHighlights** [25] val: 500 queries with a single relevant window (a seed-0 sample of the 1,020 single-window queries), videos of about 150 s. **ActivityNet Captions** [24] val_1: 500 queries, one sentence per video, videos of about 115 s. **Charades-STA** [22] test: 500 queries, videos of about 30 s. For the last three,

the question is *"When does the following happen in the video: <query>?"* and the window is the whole video. Our QVHighlights and ActivityNet copies are low-resolution re-encodes (153 and about 84 visual tokens per frame pair, against 660 on S-EMBER), so we also report budgets in absolute tokens.

Budget. 10% of the visual tokens of the 64-frame memory, timestamps included (S-EMBER: about 2,140 tokens; QVHighlights: about 520). ActivityNet is reported at 30% (400–900 tokens), because at 10% the memory holds only 120–300 tokens (Sec. 6.5).

Baselines. All tokens (no pruning); random tokens; FastVID [33], VidCom^2 [30], FlashVID [18] (vision-side stages) and SemVID [15], re-implemented and checked to keep the same tokens as the authors' code on test inputs; each with and without PCD. **[TODO: SnapKV-style observation-window eviction, PruneVid, global question top-k without quotas, and FV-Action / attention readout on the same memory.]**

Floor-adjusted evaluation. For every dataset we report three video-free or video-light references. (i) The *best constant interval* $[aq, bq]$ over a 20×20 grid, tuned on the test questions, which upper-bounds any constant rule. (ii) *Length rescaling*: the no-PCD prediction stretched about its start by a factor fitted once on S-EMBER (1.76) and applied unchanged to the other datasets; it tests whether a gain could come from longer intervals alone. (iii) The split into *closed* questions (gold end $< 0.8q$) and the rest. End-snapping on S-EMBER is discussed in Appendix C.

Metrics and statistics. mIoU and R@$\{0.3, 0.5, 0.7\}$; on S-EMBER also answer accuracy and GQ@0.5, judged with the official S-EMBER judge prompt by a local Qwen3.8-27B-FP8 model, all rows of a table in one judging session. Answers that state times as minutes:seconds are converted to seconds for every method alike (Appendix A). We use a paired bootstrap over questions (2,000 resamples) and mark with * differences whose 95% interval excludes 0. **[TODO: video-clustered bootstrap.]**

Protocol. All design choices (scores, quotas, α, scope, reserve) were made on S-EMBER. The configuration was then frozen and run once on QVHighlights, ActivityNet and Charades-STA. **[TODO: held-out S-EMBER videos as a test split.]**

6. Results

We organise the results by question. Section 6.1 compares CoCo with all baselines and floors on four datasets. Section 6.2 shows how published pruners behave on grounding. Sections 6.3 and 6.4 analyse the two components. Sections 6.5 and 6.6 study the budget and how to spend it. Section 6.7 compares with upper bounds that use the gold interval, Sec. 6.8 analyses the optional retrieval stage, Sec. 6.9 lists what did not work, and Sec. 6.10 analyses the remaining errors.

Reading the tables. Numbers are mIoU, or mIoU / R@0.5 where two are given. Δ is a paired difference; * marks differences whose 95% paired bootstrap interval excludes 0. Tables built from the early S-EMBER phases (Tabs. 11 to 14, 18 and 20 and Tab. 2) predate the answer reserve, so PCD there runs without it, and use the official parser and batch size 1; on S-EMBER the two parsers differ by at most 0.7 mIoU and batch size changes no paired result. In every table the best value is in **bold** and the second best underlined, per column unless a caption says per row; floors and differences (Δ) are not ranked.

6.1. Main results

Tables 4 and 5 lead to three conclusions.

(i) On the least biased benchmark, CoCo nearly doubles the uncompressed model. On QVHighlights, with 10% of the tokens, CoCo reaches 54.8 R@0.5 (R@0.7: 30.2) against 28.6 (18.4) for all tokens and 25.0 (16.2) for random tokens, and clears every floor by a wide margin (+22.1 mIoU over the best constant, +8.6 over length rescaling).

(ii) On S-EMBER, the gain is real where the window end is not a shortcut. CoCo beats all tokens, random selection and SemVID under the same decoding. Against the floors its pooled gain is small and not significant, because two thirds of S-EMBER intervals end at the question time. On closed intervals, where that shortcut does not exist, CoCo gains $+7.6^*$ mIoU over the best constant and $+7.5^*$ over length rescaling (Tab. 21).

(iii) On ActivityNet and Charades-STA, the base model is the bottleneck. The uncompressed model is far below the constant floor on both (22.9 *vs*. 37.9 and 16.6 *vs*. 29.7). CoCo improves on all tokens and on every pruner on ActivityNet, but no method reaches the floor, and on Charades-STA Coverage is worse than random selection (Sec. 6.6). We report these as limits, not successes.

Answers. CoCo also improves the answers (Tab. 6): accuracy 21.7 *vs*. 17.3 for random tokens ($+4.4^*$) and 19.8 for all tokens, and GQ@0.5 13.3 *vs*. 7.4 and 8.8. Coverage alone gives the best accuracy (22.5); PCD trades about one point of accuracy for its grounding gain, and differences below about two points of accuracy are within the decoding noise of the judge setup. **[TODO: SemVID + PCD (with reserve) in the same judging session.]**

Question types. CoCo improves duration and counting questions most (Tab. 7). Location questions ("where did I put …") remain essentially unsolved by every method (mIoU $\leq$ 11): 62 of the 100 location questions have annotators who disagree on the interval (mean pairwise IoU < 0.6), which caps any method.

Table 4. **Grounding with 10% of the visual tokens** (ActivityNet: 30%), mIoU / R@0.5. Grey rows are floors: the best video-free constant interval, tuned on the test questions, and the Retrieve + Coverage prediction without PCD lengthened by a factor fitted on S-EMBER. Coverage rows include optional retrieval, which the coverage gate disables on Charades-STA. Empty cells: not yet computed. S-EMBER closed intervals: Tab. 21.

Method	S-EMBER	QVHighlights	ActivityNet (30%)	Charades-STA
Best constant interval	*36.9 / 34.7*	*26.8 / 17.6*	*37.9 / 35.8*	*29.7 / 29.6*
Length rescaling of $R + C$	*38.3 / 40.0*	*40.2 / 42.4*	*34.0 / 33.0*	*24.0 / 17.6*
All tokens (no pruning)	28.0 / 26.7	29.1 / 28.6	22.9 / 18.2	16.6 / 14.2
Random tokens	25.6 / 24.6	26.9 / 25.0	22.9 / 18.6	17.4 / 13.8
FastVID [33]	17.9 / 15.8	19.2 / 15.4		13.3 / 11.6
VidCom^2 [30]	20.4 / 18.7	19.9 / 18.2		12.3 / 8.4
FlashVID [18]	22.7 / 21.5	22.8 / 20.4	20.5 / 15.4†	14.1 / 10.8
SemVID [15]	23.3 / 20.6	24.5 / 21.2	22.8 / 18.6	14.6 / 11.2
Random + PCD	35.0 / 33.9	37.8 / 39.2	31.4 / 29.4	**26.2 / 24.0**
SemVID + PCD	34.8 / 35.8	34.8 / 34.8	30.2 / 27.8	21.7 / 19.2
Coverage only	32.5 / 34.1	34.5 / 35.6	25.4 / 21.2	16.4 / 13.6
CoCo (Coverage + PCD)	**39.3 / 40.8**	**48.8 / 54.8**	**34.7 / 32.4**	23.3 / 21.2

† at 10%.

Table 5. **Paired gains of CoCo in mIoU** (*: 95% paired bootstrap interval excludes 0).

CoCo *vs.*	S-EMBER	QVHighlights	ActivityNet (30%)	Charades-STA
All tokens	+11.3 **[TODO: sig.]**	+19.7*	+11.8*	
Random + PCD	+4.2*	+11.0*	+3.3*	−2.9*
SemVID + PCD	+4.5*	+14.1*	+4.5*	
Best constant	+2.4	+22.1*	−3.1*	−6.4*
Length rescaling	+1.0	+8.6*	+0.8	−0.7

Table 6. **Grounded QA on S-EMBER** (10%; one judging session). ΔAcc. *vs.* random tokens.

Method	Acc.	GQ@0.5	mIoU	ΔAcc.
All tokens	19.8	8.8	28.0	+2.5
Random tokens	17.3	7.4	25.6	–
FastVID	15.2	4.8	17.9	−2.1
VidCom^2	16.2	5.9	20.4	−1.1
FlashVID	17.3	6.1	22.7	+0.0
SemVID	19.8	7.6	23.3	+2.5
Coverage, uniform frames	**22.5**	11.2	31.7	+5.3*
Random + PCD	20.4	12.2	35.0	+3.2
SemVID + PCD$^\circ$	17.9	9.9	32.3	+0.6
CoCo	21.7	13.3	**39.3**	+4.4*
CoCo + coverage gate	22.1	**13.5**	**39.3**	+4.8*

$^\circ$ earlier runs: PCD without its answer reserve.

Table 7. **S-EMBER by question type**, mIoU / R@0.5 (10%).

Method	Duration (191)	Counting (184)	Location (100)
All tokens	37.6 / 40.3	30.1 / 25.5	5.9 / 3.0
Random tokens	33.2 / 35.1	29.6 / 26.1	3.5 / 2.0
Coverage	43.2 / 46.6	35.3 / 36.4	7.1 / 6.0
Random + PCD	41.0 / 40.8	41.9 / 40.2	**11.0 / 9.0**
SemVID + PCD	38.9 / 36.6	44.6 / **50.0**	8.9 / 8.0
CoCo	**47.6 / 50.8**	**46.9** / 48.4	9.3 / 8.0

6.2. QA pruners on grounding

Every published pruner (Fig. 4) falls below random token selection on S-EMBER (−2.3 to −7.6 mIoU, all significant), QVHighlights (−2.4 to −7.7; SemVID not significant) and Charades-STA (−2.8 to −5.2, all significant); on ActivityNet at 10% the input-level pruners are above random, which we discuss in Sec. 6.6. Random selection varies little across seeds (26.0 ± 0.6 mIoU over three seeds on S-EMBER), so the gaps are not noise. The ordering follows coverage (Tab. 1): the methods that empty frames most aggressively lose most, and those that spread tokens lose least. Accuracy, in contrast, is barely affected (Tab. 6): FastVID's accuracy is within noise of random selection (−2.1, n.s.) while its mIoU is 7.6 lower. A pruner built to keep answers keeps answers, not time. PCD helps every pruner (+7 to +12 mIoU) but does not close the gap to CoCo.

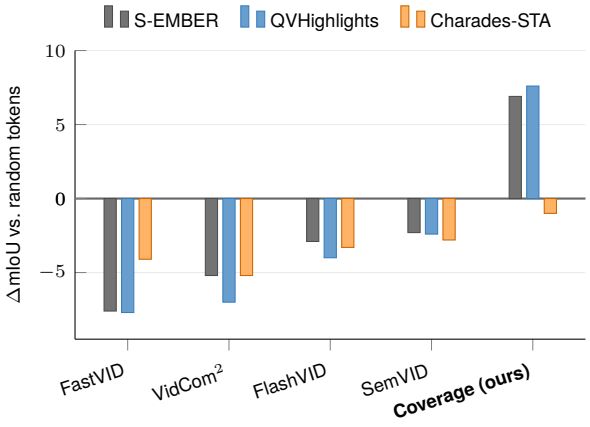


Figure 4. **QA pruners fall below random selection on grounding.** Paired mIoU difference to random token selection at the same 10% budget, greedy decoding. All four published pruners lose on all three datasets (all significant except SemVID on QVHighlights). Coverage gains on the two long-video datasets and is level with random selection on 30 s Charades videos.

Table 8. **Component ladder** (mIoU / R@0.5; ActivityNet at 30%). R: retrieval frames; C: Coverage; PCD: prior-contrastive decoding.

Configuration	S-EMBER	QVHighlights	ActivityNet
Uniform, random tokens	25.6 / 24.6	26.9 / 25.0	22.9 / 18.6
R, random tokens	29.4 / 29.5	28.7 / 28.0	–
C (uniform frames)	31.7 / 32.4	33.7 / 33.0	24.6 / 21.6
R + C	32.5 / 34.1	34.5 / 35.6	25.4 / 21.2
R + C + PCD (CoCo)	**39.3 / 40.8**	**48.8 / 54.8**	**34.7 / 32.4**

6.3. Analysis of Coverage

Contribution. Table 8 adds the components one at a time. Coverage is the largest gain on the memory side: $+6.1^*$ mIoU on S-EMBER and $+6.8^*$ on QVHighlights over random tokens. Retrieval adds $+0.8$ to $+0.9$ on top (not significant on any dataset); once tokens are chosen by the question, which frames are encoded matters less. **[TODO: Coverage + PCD without retrieval on all datasets.]**

Design choices. Table 9 varies one choice at a time. *The quota should be equal*: five relevance-weighted variants, from SigLIP similarity, attention mass or both, with floors of 25–50%, are within ± 0.5 mIoU of the equal quota and none is significant (the uniform-frame SigLIP variant was our pre-registered primary test). Quotas that follow raw attention mass are pulled by attention sinks and recency bias: a single frame received up to 227 of about 2,140 kept tokens. *The score should be the actual question*: HERMES's generic probe questions give 23.7 even with an equal quota, and 18.3 without. *Deterministic top-k beats sampling* from a coverage-floored density (31.7 *vs.* 30.0); with the quota in place, coverage is already guaranteed. A contrastive token score that subtracts the attention of a generic sentence, meant to cancel sinks, does not help.

Table 9. **Design choices of Coverage** (S-EMBER, greedy decoding). Frames: R = retrieval, U = uniform. Δ *vs.* the equal quota with question top-k on the same frames.

Token selection	Frames	mIoU	Δ
Equal quota, question top-k	R	32.5	–
Quota ∝ SigLIP relevance (floor 50%)	R	**33.0**	+0.5
Quota ∝ SigLIP (floor 25%)	R	32.6	+0.1
Quota ∝ SigLIP + attention	R	32.9	+0.4
Quota ∝ attention mass	R	32.1	−0.4
Contrastive token score‡	R	**33.0**	+0.5, n.s.
One token set shared by all layers	R	31.9	−0.6
Equal quota, question top-k	U	31.7	–
Quota ∝ SigLIP relevance	U	31.4	−0.2
Equal quota, sampling instead of top-k	U	30.0	−1.7
Contrastive token score‡	U	30.5	−1.2
HERMES score, equal quota (stratified)	U	23.7	
HERMES score, global top-k	U	18.3	
Question, global top-k, no quota	U	**[TODO:]**	

‡ $\log a(\text{question}) - \log a(\text{generic sentence})$.

Table 10. **Per-layer *vs*. shared token sets** (S-EMBER, 10%). Δ = shared − per layer.

Method	Per layer	Shared	ΔmIoU
Random tokens	25.6	26.3	+0.7
Random + PCD	35.0	35.8	+0.7
R + C	32.5	31.9	−0.6
CoCo	**39.3**	**37.8**	-1.5^*

Per-layer selection. Table 10 compares one token set per layer with a single set shared by all layers. Sharing does not change random selection, which rules out per-layer diversity as the reason random selection beats input-level pruners. It costs CoCo -1.5^* mIoU: different layers attend to different tokens, and question-driven selection benefits from letting each choose.

6.4. Analysis of PCD

PCD is a plug-in for any memory. Table 11 applies PCD to eight memories: three random budgets, the uncompressed model, HERMES and stratified pruning, and two HERMES streaming memories. It improves mIoU significantly on all eight ($+4.6$ to $+8.0$) and GQ@0.5 on six, roughly halves the early starts everywhere, and its gains are largest where the memory is smallest. With 10% of the tokens and random selection it already beats the uncompressed model (32.6 *vs.* 28.0). PCD also transfers to FastVID ($+7.6^*$ mIoU, leak 38% → 12%), the pruner that loses coverage most.

The no-video counterfactual is the right one. Table 12 changes only what the memory is contrasted against. The no-video input isolates the prior and removes it (early starts

Table 11. **PCD on eight memories** (S-EMBER, Acc. / mIoU / GQ@0.5; earlier runs: PCD without its answer reserve). Plain PCD: on all tokens with constant α=0.5; PCD: confidence-adaptive strength on time values only (ours). Δ PCD *vs*. greedy. Best and second best per row.

Memory	Greedy	Plain PCD	PCD	ΔmIoU	ΔGQ@0.5	Early starts
Random 5%	15.8 / 23.6 / 5.9	15.4 / 29.2 / 9.1	**16.4 / 31.5 / 9.9**	+8.0*	+4.0*	39% → 20%
Random 10%	15.2 / 25.8 / 7.2	**16.2** / 32.1 / **10.1**	**16.2 / 32.6 / 10.1**	+6.8*	+2.9	38% → 19%
Random 25%	17.7 / 28.2 / 8.2	**18.9** / 33.0 / 11.8	**18.9 / 34.9 / 13.3**	+6.7*	+5.1*	37% → 21%
All tokens	18.5 / 28.0 / 8.6	19.8 / 31.5 / 11.2	**21.5 / 32.7 / 13.3**	+4.6*	+4.6*	41% → 23%
HERMES 10%	13.1 / 18.3 / 4.0	**15.4** / 22.6 / **7.2**	13.3 / **24.1** / 5.9	+5.7*	+1.9	56% → 31%
Stratified 10%	14.1 / 23.7 / 6.1	14.5 / 29.9 / 8.8	**15.8 / 31.0 / 10.5**	+7.3*	+4.4*	36% → 17%
HERMES streaming 4k	13.5 / 21.6 / 4.0	**17.1** / 27.4 / **8.2**	15.4 / **28.6** / **8.2**	+6.9*	+4.2*	45% → 24%
HERMES streaming 6k	14.3 / 22.6 / 4.8	14.7 / 28.1 / 7.6	**15.4 / 29.8 / 8.4**	+7.2*	+3.6*	46% → 24%
Mean of 8	15.3 / 24.0 / 6.1	16.5 / 29.2 / 9.3	**16.6 / 30.7 / 10.0**	+6.7	+3.9	

Table 12. **What to contrast against** (S-EMBER, random 10%). Only the counterfactual input changes; the main input is always the real memory.

Counterfactual	mIoU	GQ@0.5	Early
None (greedy)	25.8	7.2	38%
No video (ours)	**32.1** (+6.2*)	**10.1** (+2.9*)	**19%**
Noised frames (VCD [26])	26.3 (+0.5)	7.8	30%
Permuted timestamps	+0.9 (n.s.)	–	–
Evidence-against rule, α=1 / 2	26.5 / 27.3	–	–
Attention gain (PAI-style)	+1.3 (n.s.)	+0.8	42%
Instruction "do not assume early"	25.7	6.9	37%

Table 13. **Time scope and adaptive strength** (S-EMBER, Acc. / mIoU / GQ@0.5). Top: PCD on all tokens *vs*. time values only. Bottom: the 78 questions whose evidence truly starts in the first 5% of the window. Top: best per row.

Memory	All tokens	Time values only
Random 10%	**16.2** / 32.1 / 10.1	16.0 / 32.1 / **10.3**
Random 25%	**18.9** / 33.0 / 11.8	17.7 / **34.0** / 11.8
All tokens	19.8 / 31.5 / 11.2	**20.8 / 32.9 / 12.4**
HERMES 10%	**15.4** / 22.6 / 7.2	14.7 / **24.3** / **7.4**
Streaming 4k	**17.1** / 27.4 / 8.2	16.2 / **29.2** / **9.3**
Early-evidence questions (78), all tokens memory		
Greedy	21.8 / 49.2 / 16.7	
PCD, constant α	16.7 / 55.3 / 12.8	
PCD, adaptive α	**23.1 / 56.9 / 19.2**	

Table 14. **Decision rules replayed at the first time value** (S-EMBER, random 10%). Share of answers starting at "0".

Rule	Early evidence ↑	Other (leak) ↓
Greedy	**69%**	16%
PCD, α=0.5	46%	**6%**
PCD, adaptive	51%	7%
Evidence-against, α=4	51%	9%
Skewed or tempered prior	50–55%	8–11%
Best threshold (uses gold)	46%	**6%**

38% → 19%). Noised frames, as in VCD, still show part of the video *and* still produce the prior, so the two runs disagree mainly about content and little of the prior is removed (+0.5 mIoU). Permuted timestamps leave the prior in place as well. A one-sided rule that only penalises tokens the video argues against is worse than PCD (-5.6^* and -4.8^*): without video the model is about 99% sure of "0", so even a well-supported "0" looks like evidence against. Boosting visual attention during decoding, or stating the bias in the instruction, leaves the prior in place.

Time scope and adaptive strength. Contrasting every token also changes words that have nothing to do with time: the answer text changed in 85% of questions, lists of more than eight moments rose from 12 to 56, and over-counting rose from 14% to 34% of counting questions. Restricting PCD to time values keeps the grounding gain and disturbs the text less (Tab. 13, top). PCD also risks hurting questions whose evidence really starts at 0 s. With a constant strength, their accuracy on the uncompressed memory falls from 21.8 to 16.7; with the adaptive strength, which fades when the model is confident, it rises to 23.1 (Tab. 13, bottom; $+6.4^*$ *vs*. constant). The adaptive rule also gives the best *discrimination*, early starts on early-evidence questions minus early starts on the others (56.4 *vs*. 49.6 for greedy, all-tokens memory).

PCD is near the limit of decoding. We logged both distributions at the first time value of every answer and replayed other decision rules offline (Tab. 14). Even a threshold tuned on the gold labels cannot beat PCD's trade-off between keeping true early starts and removing leaked ones. The remaining early-start errors are cases where the memory itself is unsure; they need evidence, not a better decoding rule.

Table 15. **The answer reserve inside PCD**: PCD *vs*. PCD without the reserve (paired ΔmIoU), and unparseable answers without → with it.

Configuration	S-EMBER	QVHighlights	Charades
Random + PCD	$+1.9^*$	$+1.5^*$	$+2.7^*$
SemVID + PCD	$+2.5^*$	–	$+5.6^*$
R + C + PCD	$+1.8^*$	$+6.1^*$	$+2.4^*$
Unparseable (of 500)	26–39 → 0	25–60 → 0	28–68 → 0

Is PCD more than a longer interval? PCD lengthens predicted intervals (median predicted/gold length 0.57 → 0.67 on S-EMBER), and longer intervals alone raise mIoU on benchmarks with long gold intervals. The length-rescaling floor tests this directly (Tab. 5, last row). On QVHighlights ($+8.6^*$) and on S-EMBER closed intervals ($+7.5^*$, Tab. 21), CoCo clearly beats rescaling: PCD moves the interval toward the evidence, not just outward. On pooled S-EMBER, ActivityNet and Charades-STA the two are statistically tied, so on those sets the benefit of PCD cannot be separated from a calibrated lengthening. **[TODO: start/end error decomposition on closed questions; mismatched-video and shuffled-frame negatives.]**

Answer reserve. PCD rewards fine time steps that the prior finds unlikely, so it sometimes lists one continuing activity at one-second steps until the token limit. The reserve removes every such failure and improves every PCD configuration we tested (Tab. 15), most where run-ons were frequent (SemVID + PCD on Charades, CoCo on QVHighlights). It costs nothing when the model answers within the limit, which every greedy run does.

Prompt format. PCD needs a reasoning-style answer: with the official direct-answer prompt it does not help (+0.4 mIoU). An *event* timeline that lists start and end per occurrence (`Event: 12.0 -- 15.5 seconds`) improves the greedy baseline by $+7.4^*$ mIoU by stating extents, and PCD still adds +2.1 to +3.3 on top (35.0 mIoU, 12.6 GQ@0.5 at 768 tokens). The two fixes overlap: the format corrects extents, PCD corrects the leak. CoCo uses the simpler timeline in all main results. **[TODO: event timeline with CoCo.]**

6.5. Budget

Table 16 and Fig. 5 vary the budget at 64 frames and show two regimes. Above roughly 500 kept tokens, question-driven selection is flat or even improves as the budget *shrinks*: on S-EMBER, Retrieve + Coverage rises from 31.0 (20%) to 34.6 (5%) and CoCo from 36.8 to 39.3, while random selection and SemVID degrade. Pruning by the question removes distractors as well as cost: at 5% CoCo leads SemVID + PCD by $+7.4^*$ mIoU, at 20% by +1.5 (n.s.). Below roughly 300 kept tokens (QVHighlights 5%, ActivityNet 10%), the equal quota leaves only 2–4 tokens per frame pair, the top-attended tokens carry little content, and CoCo collapses (9.7 and 11.6), below random selection and below SemVID, whose cross-frame merging degrades more gracefully. The threshold is absolute, not relative: it depends on the number of tokens per frame, which is low in our re-encoded QVHighlights and ActivityNet copies.

6.6. Fewer frames or fewer tokens per frame?

The collapse at small budgets is not a limit of the equal quota itself but of how the budget is spent. In Tab. 17 every row of a block keeps the same number of tokens and only the split between frames and tokens per frame changes. With 32 frames instead of 64, each frame pair keeps twice as many tokens, and CoCo recovers from 9.7 to 43.1 mIoU on QVHighlights and from 11.6 to 33.9 on ActivityNet, where it now beats SemVID + PCD (+5.5) and nearly matches its own 64-frame result at three times the budget (34.7). Halving again to 16 frames loses a little: past a point, temporal coverage becomes the bottleneck. The baselines gain from fewer frames too, but less. On Charades-STA (30 s videos) fewer frames do not help any method. We therefore recommend the rule of Sec. 4.1: keep at least about eight tokens per frame pair and reduce F when the budget does not allow it. At equal tokens, more frames with fewer tokens each does not help either: 128 frames at 5% on S-EMBER give +1.2 mIoU (n.s.) without PCD and -3.6^* with it, with 74–78 unparseable answers.

6.7. Comparison with gold-informed upper bounds

Table 18 gives the memory information it should not have. At the same budget, PCD without any gold information beats perfect token selection on grounding ($+4.7^*$ mIoU over the oracle selector at 10%, $+4.5^*$ at 5%): knowing which tokens to keep does not stop the prior, and the oracle selector still leaks 33% of the time. Against the strongest oracle, which sees only the evidence, PCD closes most of the grounding gap (GQ@0.5 13.3 *vs*. 13.9, n.s.), and the remaining gap is in the answers (accuracy 21.5 *vs*. 26.3^*). Even the oracle window leaks 45%: with every frame inside the evidence, the model still often starts near 0 s. More evidence alone does not remove the prior; the two levers are complementary.

6.8. Analysis of the optional retrieval

The SigLIP 2 index does find the evidence: over the seconds before each question, the AUROC of evidence seconds against the rest is 0.663, and the top-ranked second lies in the gold interval for 63% of questions (chance 36%) and 41% of location questions (chance 13.5%). Retrieval-guided frames put 40% of the memory inside the evidence, against 36% for uniform frames. This helps random tokens ($+3.8^*$ mIoU) but adds only +0.8 to +0.9 (n.s.) once tokens are chosen by the question, because both stages focus the memory on the

Table 16. **Budget sweep at 64 frames** (mIoU). The second row gives the number of kept tokens per memory.

	S-EMBER			QVHighlights			ActivityNet		
Budget	5%	10%	20%	5%	10%	20%	10%	30%	50%
Kept tokens	∼1,070	∼2,140	∼4,280	∼260	∼520	∼1,040	120–300	400–900	600–1,500
Random	23.9	25.6	27.4		26.9		17.8	22.9	23.5
SemVID	20.9	23.3	26.5		24.5		21.2	22.8	22.3
R + C (no PCD)	34.6	32.5	31.0		34.5		11.3	25.4	25.0
SemVID + PCD	31.9	34.8	35.3		34.8		**26.8**	30.2	31.4
Random + PCD	34.5	35.0	36.5		37.8		21.5	31.4	32.2
CoCo	**39.3**	**39.3**	**36.8**	9.7	**48.8**	49.4	11.6	**34.7**	**34.4**

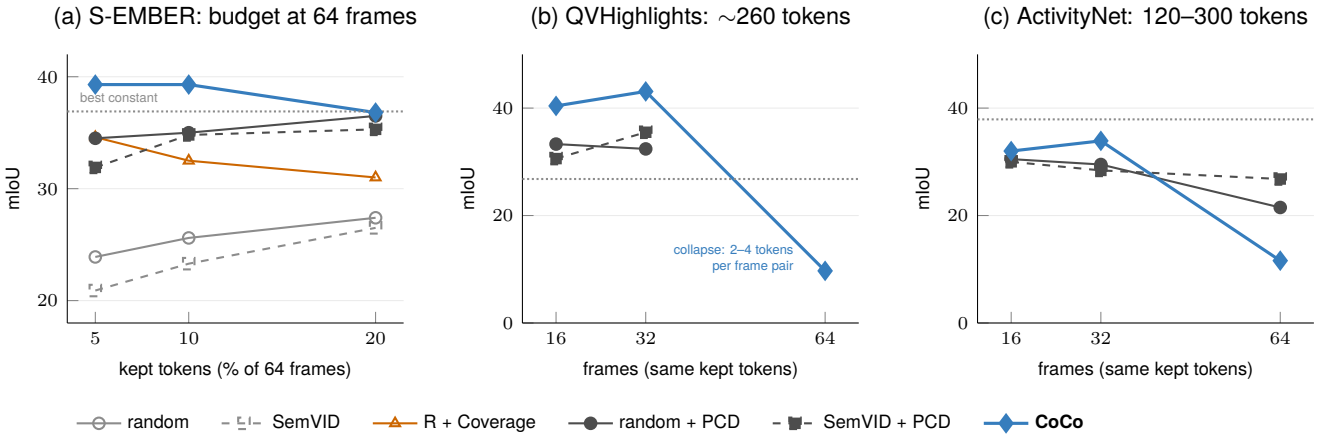


Figure 5. **Budget and how to spend it. (a)** At 64 frames, question-driven selection improves as the budget shrinks while random selection and SemVID degrade; CoCo leads at every budget. **(b, c)** At a fixed, very small number of kept tokens, 64 frames leave 2–4 tokens per frame pair and CoCo collapses; spending the same tokens on 32 frames recovers it (QVHighlights 9.7 → 43.1, ActivityNet 11.6 → 33.9). Dotted grey: best video-free constant.

question. Subtracting a question-free prior from SigLIP's score does not help (AUROC 0.635–0.666), as expected for a retriever trained with a contrastive loss, whose score already estimates pointwise mutual information [36]. Re-weighting the token budget of uniform frames by the same relevance adds +1.3 (n.s.): which frames are encoded matters more than how the budget is split among fixed frames.

6.9. What did not work

Table 19 collects the ideas that gave little or nothing. They share a lesson: at a fixed budget, redistributing evidence (zooming, re-weighting frames, unequal quotas) is close to its ceiling, and only removing the prior where it acts (PCD) or guaranteeing coverage (the equal quota) gives large gains.

6.10. Where answers still fail

Table 20 shows that PCD works where it was aimed: intervals entirely before the gold one, the signature of the prior, fall from the largest error type to less than a fifth of answers. The remaining failures point to four limits. (i) *Answers, not intervals*: twice as many answers are well grounded but wrong as grounded and right; counting answers do not use the events the model itself listed, and an IoU of 0.5 still allows a two-fold duration error. (ii) *Recent events*: 81% of the intervals that still end before the gold one have evidence in the second half of the window, where an earlier similar event is picked; GQ@0.5 is 4.4 when the evidence lies in the last third of the window. (iii) *Sparse evidence*: with fewer than 4 of the 64 frames inside the evidence, GQ@0.5 is 1–2% for any method, against 15% with 16 or more; no decoding rule recovers evidence that is not in the input. (iv) *Label ambiguity*: on the 98 questions whose annotators disagree (mean pairwise IoU < 0.6), every method scores near zero.

7. Limitations

Base-model floors. On ActivityNet Captions and Charades-STA, Qwen3-VL-8B with 64 frames stays below the best constant interval even without pruning, and so does every compressed variant. Our claims on these datasets are relative to the model.

Minimum tokens per frame. Equal quotas fail when a

Table 17. **Spending a fixed token budget on fewer frames** (mIoU / R@0.5, all with PCD). Within each block every row keeps the same number of tokens: F frames at keep ratio $\rho \cdot 64/F$. Best and second best per row.

Dataset (kept tokens)	Frames × keep ratio	Random + PCD	SemVID + PCD	CoCo
QVHighlights (∼260)	64 × 5%	–	–	9.7 / 6.6
	32 × 10%	32.4 / 29.2	35.5 / 33.6	**43.1 / 46.6**
	16 × 20%	33.3 / 32.4	30.6 / 28.0	**40.4 / 41.0**
ActivityNet (120–300)	64 × 10%	21.5 / 18.2	**26.8 / 23.6**	11.6 / 7.6
	32 × 20%	29.5 / 26.8	28.4 / 25.8	**33.9 / 32.0**
	16 × 40%	30.5 / 26.8	30.0 / 27.6	**32.0 / 29.8**
Charades-STA (∼10% of 64)	64 × 10%	**26.2 / 24.0**	21.7 / 19.2	23.3 / 21.2
	32 × 20%	**24.6 / 22.0**	20.8 / 16.0	23.6 / 21.0[§]

[§] Coverage on uniform frames; retrieval is gated off on Charades-STA.

Table 18. **Upper bounds that use the gold interval** (S-EMBER, greedy unless noted; earlier runs: PCD without its answer reserve). Oracle selector: the gold frames' tokens are kept first. Oracle window: all 64 frames inside the gold interval.

Memory	Acc.	mIoU	R@0.5	GQ	Leak
No video	10.5	7.9	4.8	0.8	88%
Random 10%	15.2	25.8	24.2	7.2	30%
Random 10% + PCD	16.2	32.6	33.1	10.1	**11%**
Oracle selector 10%	17.5	27.9	25.7	8.4	33%
Gold-window zoom 10%	14.5	28.2	26.1	7.8	25%
All tokens	18.5	28.0	26.7	8.6	33%
All tokens + PCD	21.5	32.7	34.1	13.3	13%
Oracle window	**26.3**	**35.3**	**36.2**	**13.9**	45%

Table 19. **Negative results** (S-EMBER; ΔmIoU against the stated reference).

Idea	Reference	ΔmIoU
VCD-style noised-frame contrast	greedy	+0.5
Permuted-timestamp contrast	greedy	+0.9
Evidence-against decoding rule	PCD	−5.6*
Attention gain while decoding	greedy	+1.3
Bias stated in the instruction	greedy	−0.1
Relevance-guided frame zoom	greedy	−1.5
Zoom inside the gold window	greedy	+2.3
Unequal per-frame quotas (5 variants)	equal quota	−0.4 to +0.5
Contrastive token score	question top-k	−1.2 to +0.5
Prior-subtracted retrieval	SigLIP	AUROC ≤ +0.003
128 frames at the same budget	64 frames	+1.2 / −3.6*

Table 20. **Error types** on S-EMBER, pooled over eight memories (3,800 answers), greedy *vs.* PCD (earlier runs: PCD without its answer reserve).

Outcome	Greedy	PCD	Δ
IoU ≥ 0.5, answer correct	6.1%	9.9%	+3.8*
IoU ≥ 0.5, answer wrong	16.3%	20.7%	+4.4*
Partial overlap (IoU < 0.5)	17.0%	15.7%	−1.3
Inside gold, too short	22.2%	18.5%	−3.7*
Entirely *before* gold	33.1%	18.9%	−14.2*
Entirely after gold	2.7%	5.7%	+3.0*
Covers gold, too long	2.2%	2.5%	+0.3
No answer / unparsed[¶]	0.5%	8.1%	+7.6*

[¶] removed by the answer reserve (Tab. 15).

frame pair keeps only 2–4 tokens. Spending the budget on fewer frames avoids this (Sec. 6.6), but the rule we use (at least about eight tokens per frame pair) was derived from the same results and still needs held-out validation.

Short, densely sampled videos. When uniform frames are less than a second apart, spreading tokens at random beats question-driven selection.

Not a streaming memory. The memory is rebuilt per question from the frames before the question. It is causal, but the full memory is prefilled once per question, so peak memory and time-to-first-token are those of the uncompressed model: CoCo reduces the memory the decoder reads, not the cost of building it. PCD also adds a second, prompt-only decoding pass for time-value tokens. **[TODO: cost table; write-once memory re-ranked at question time.]**

Scope of the evaluation. All results use one model, 475–500-question subsets, and a local judge with the official prompt. Every hyperparameter was chosen on the S-EMBER questions; the other datasets were run once with the frozen configuration, but a held-out S-EMBER split is still to be evaluated.

8. Conclusion

Temporal grounding asks a compressed video memory for something multiple-choice QA does not: a complete timeline. Pruners designed for QA remove parts of it and fall below random selection, and the decoder's video-free prior fills the gaps, mostly by ending intervals too early. CoCo keeps every time anchor with an equal, question-filled quota

and writes time values against the model's own no-video prediction. With a tenth of the visual tokens it grounds better than the uncompressed model on S-EMBER and QVHighlights, and it clears video-free floors wherever the base model does. The floors are a lesson of their own: on two standard benchmarks a constant interval beats the uncompressed model, and grounding results under compression should be read against them.

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A. Parsing times written as minutes:seconds

On videos longer than a minute, the model sometimes writes times as minutes:seconds, *e.g.* `Time: [1:45, 2:07]`, which the official S-EMBER parser scores as missing. We convert `h:mm:ss` and `m:ss` to seconds for every method alike. Method rankings are unchanged with the official parser. The main numbers with the official parser are: QVHighlights, CoCo 44.6 / 50.2 (*vs*. 48.8 / 54.8); ActivityNet at 30%, 33.3 / 31.0 (*vs*. 34.7 / 32.4). **[TODO: full table with the official parser.]**

B. S-EMBER closed intervals

Two thirds of S-EMBER gold intervals end at, or close to, the question time, so a constant interval ending at q is a strong floor on the pooled set. Table 21 restricts the main comparison to the *closed* questions (gold end $< 0.8q$, 33% of the questions), where that shortcut does not exist; the floors are re-tuned on this subset.

Table 21. **S-EMBER closed intervals** (10%, mIoU / R@0.5; settings as in Tab. 4). Right: paired ΔmIoU of CoCo (*: 95% paired bootstrap interval excludes 0).

Method	mIoU / R@0.5	CoCo *vs.*
Best constant interval	*32.1 / 29.9*	+7.6*
Length rescaling of $R + C$	*32.3 / 33.1*	+7.5*
All tokens (no pruning)	30.0 / 27.4	
Random tokens	28.3 / 27.4	
Random + PCD	36.8 / 38.2	
SemVID + PCD	34.3 / 34.4	
Coverage only	33.0 / 35.7	
CoCo (Coverage + PCD)	**39.7 / 43.3**	–

C. End-snapping on S-EMBER

Setting every predicted end to the question time q raises every S-EMBER method to 42–46 mIoU, more than the 39.3 of CoCo, because 67% of the gold intervals end within the last fifth of the window. On closed intervals, end-snapping *hurts* every method (to 22–24 mIoU) while CoCo keeps 39.7; on QVHighlights it lowers CoCo from 48.8 to 40.5. A decoder that decides whether an interval is still open at question time is a natural extension. **[TODO: open-interval decoding.]**

D. Reproduction

CoCo corresponds to the following overrides of the HERMES code base:

```
model=qwen3_vl_8b experiment=sember_grounding_uniform run.uniform_num_frames=64
dataset.grounding_prompt=timeline run.keep_time_tokens=surviving run.max_new_tokens=384
run.batch_size=8 run.prune_score=core_topk run.offline_keep_ratio=0.1
run.contrastive_mode=blind run.contrastive_alpha=1.0 run.contrastive_adaptive=true
run.contrastive_scope=time run.answer_reserve=64
(optional retrieval) run.retrieval=frames_cont run.retrieval_min_gap=1.0
```

Floors and paired intervals are computed by `docs/paper/analysis/floors.py`.